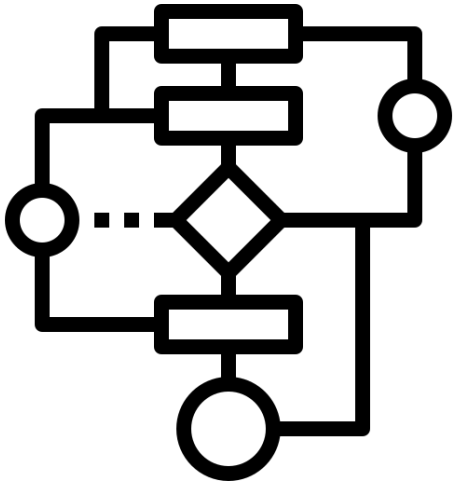
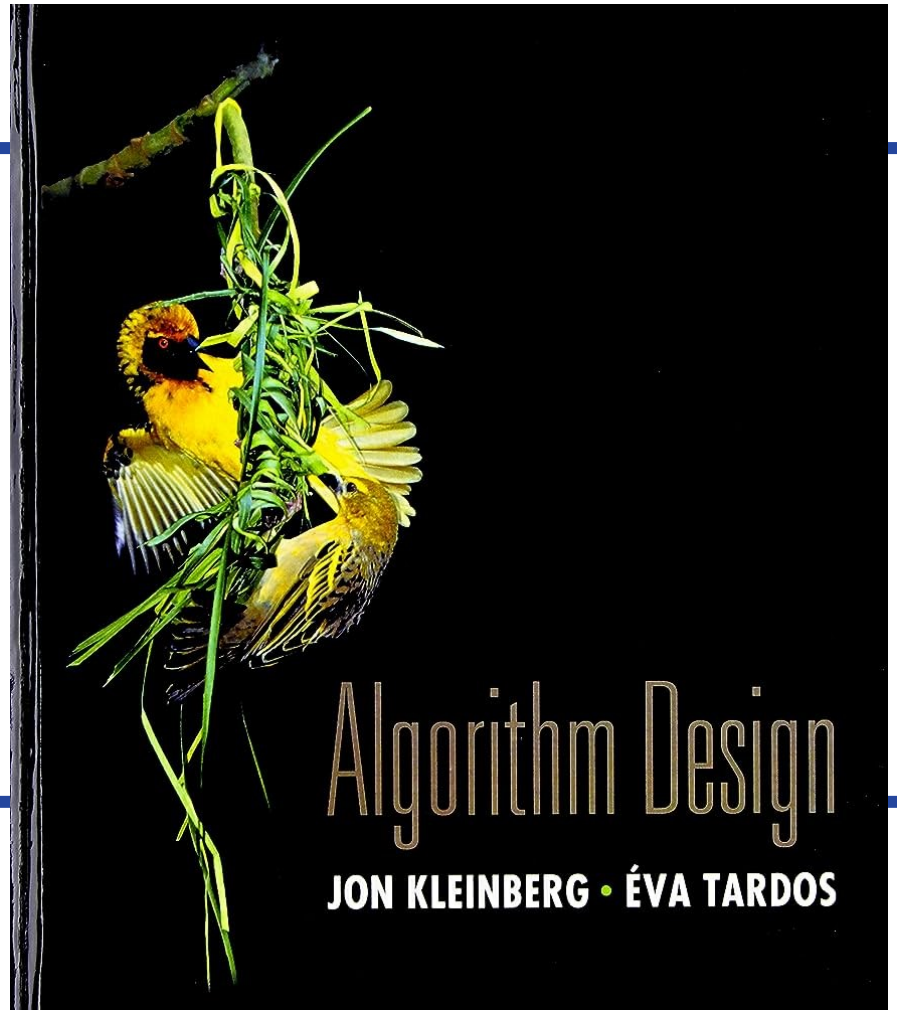


CS 310: Algorithms

Lecture 6

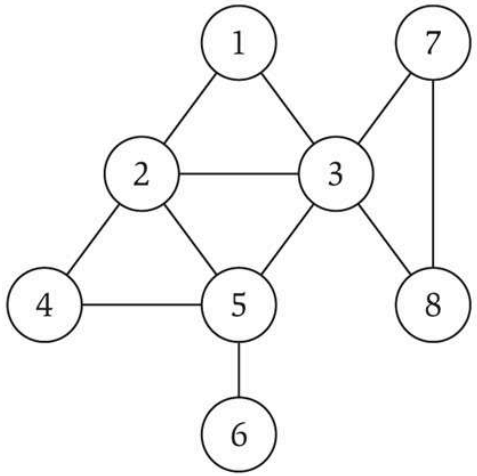
Instructor: Naveed Anwar Bhatti



Chapter 3:
Graphs

Section 3.2:
Graph Traversal

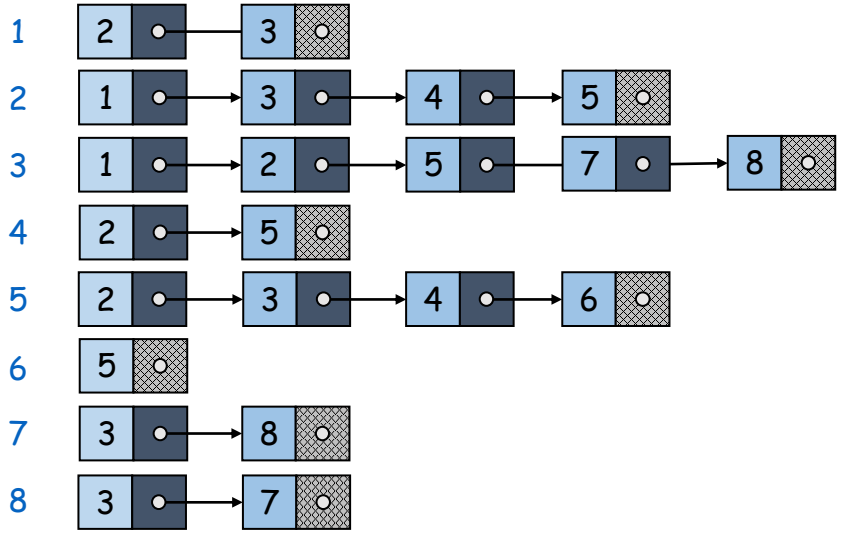
Last time: Graph Representation



Adjacency Matrix

	1	2	3	4	5	6	7	8
1	0	1	1	0	0	0	0	0
2	1	0	1	1	1	0	0	0
3	1	1	0	0	1	0	1	1
4	0	1	0	0	1	0	0	0
5	0	1	1	1	0	1	0	0
6	0	0	0	0	1	0	0	0
7	0	0	1	0	0	0	0	1
8	0	0	1	0	0	0	1	0

Adjacency List

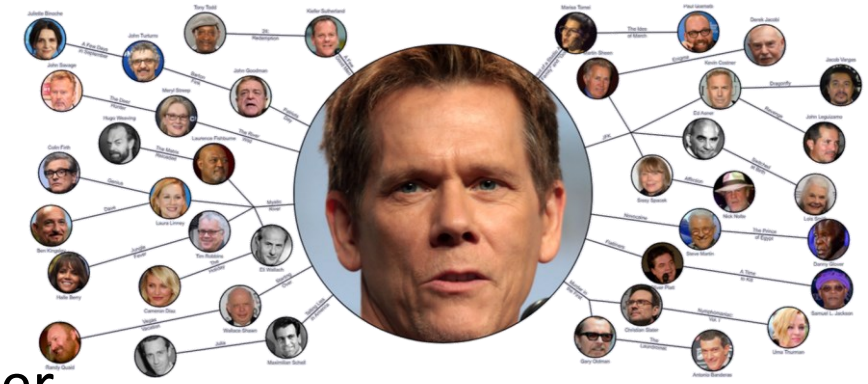


Connectivity

- **s-t connectivity problem**: Given two node **s** and **t**, is there a path between **s** and **t**?

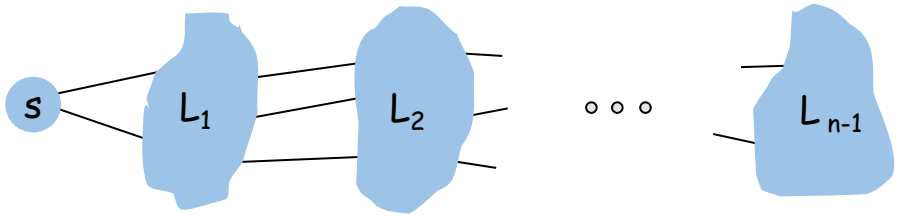
- **s-t shortest path problem:** Given two node s and t, what is the length of the shortest path between s and t?

- Applications.
  - Google Maps
  - Maze traversal
  - Kevin Bacon number
  - Fewest number of hops in a communication network



Breadth First Search

- BFS intuition: Explore outward from ***s*** in all possible directions, adding nodes **one "layer" at a time**.



- BFS algorithm:
  - $L_0 = \{ s \}$.
  - L_1 = all neighbors of L_0.
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Theorem: For each ***i***, $\mathbf{L_i}$ consists of all nodes at distance exactly ***i*** from **s**. There is a path from **s** to **t** iff **t** appears in some layer.

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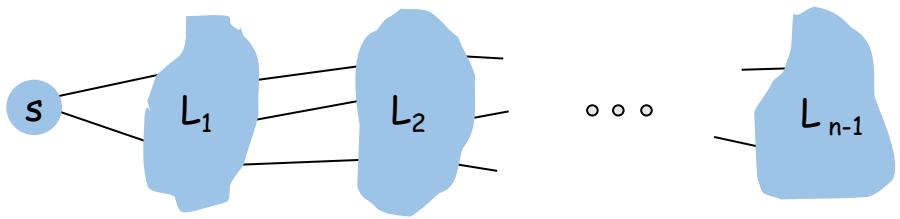

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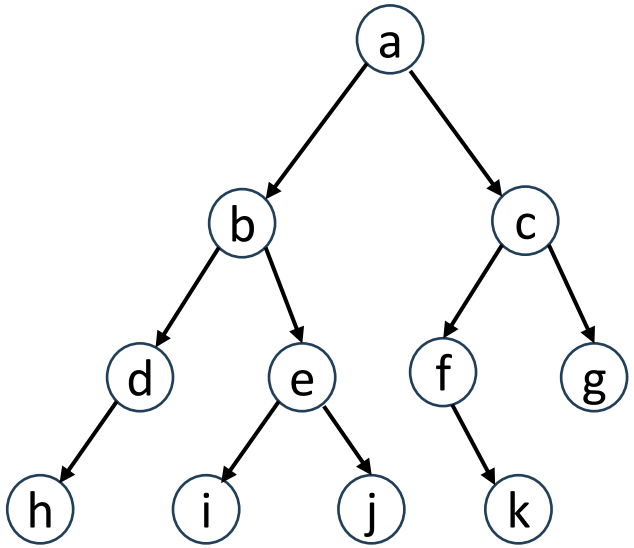

Breath First Search

7

Breadth First Search

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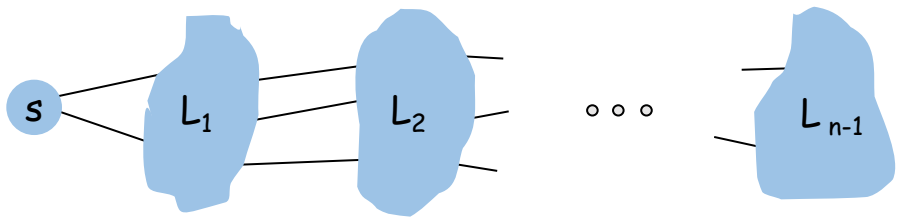

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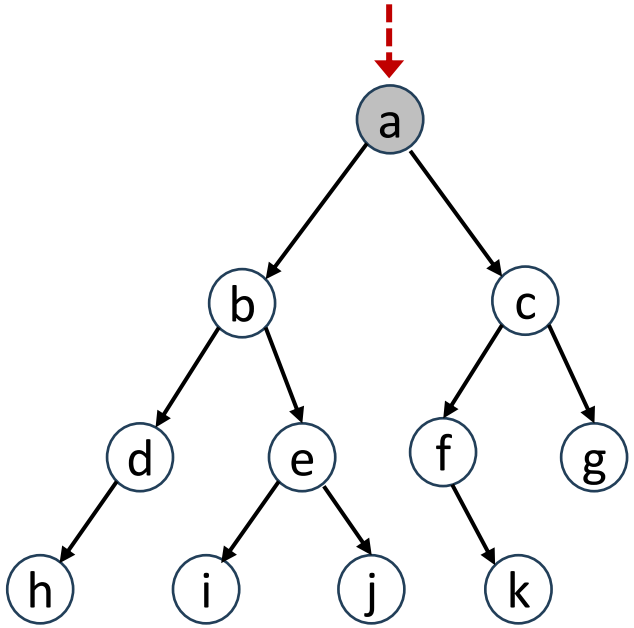

Breath First Search

8

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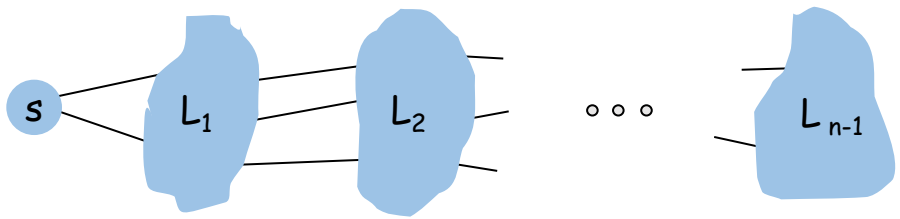

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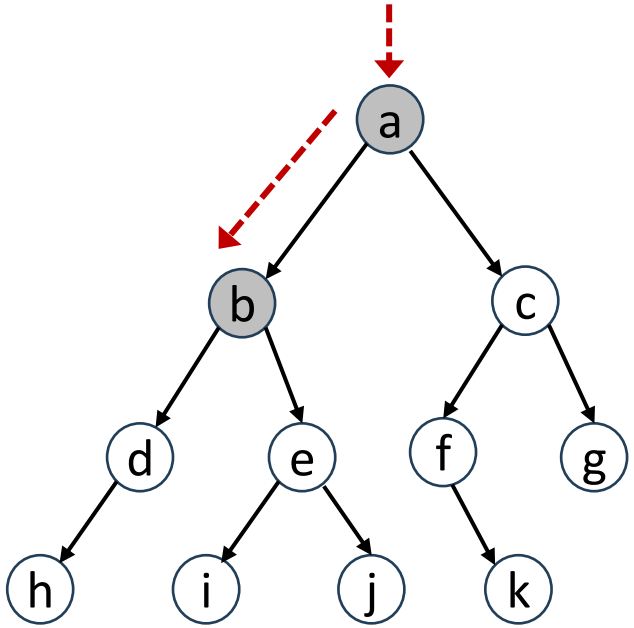

Breath First Search

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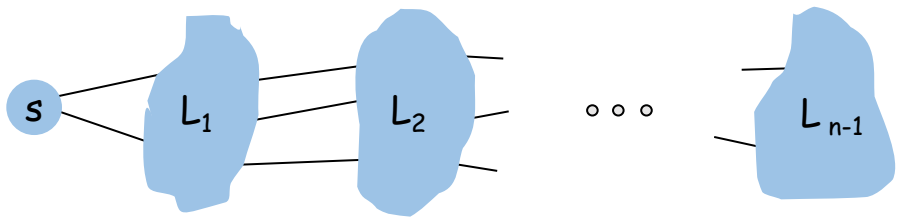

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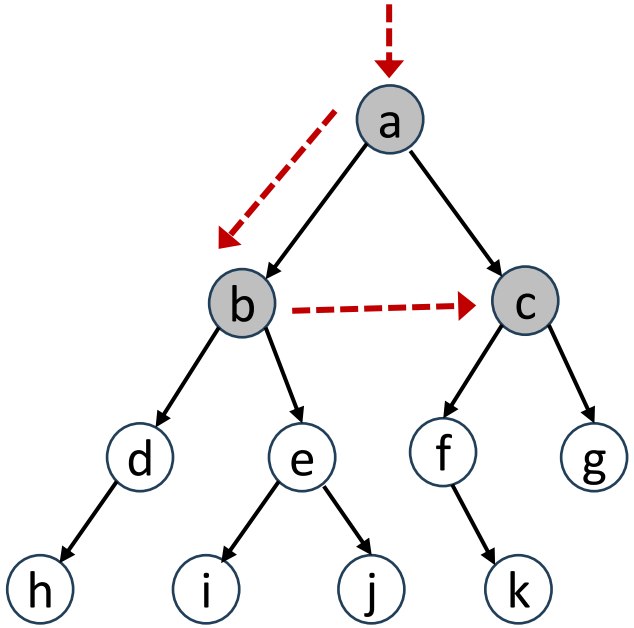

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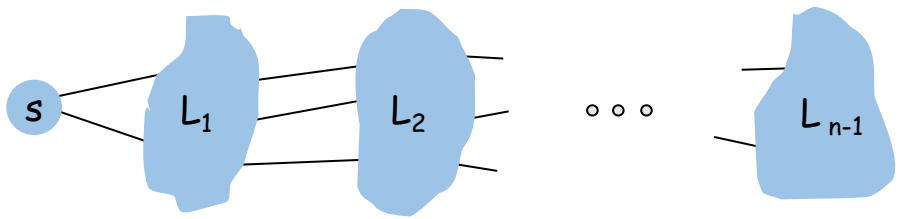

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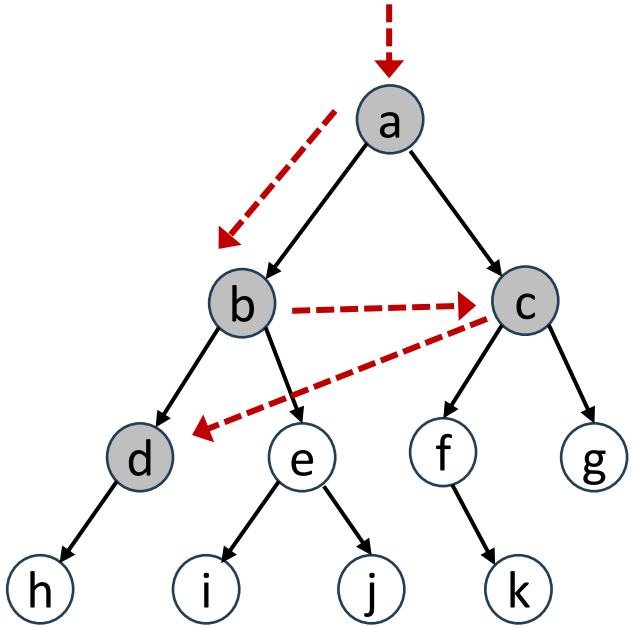

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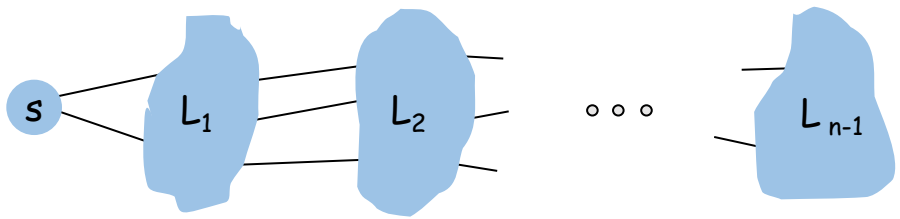

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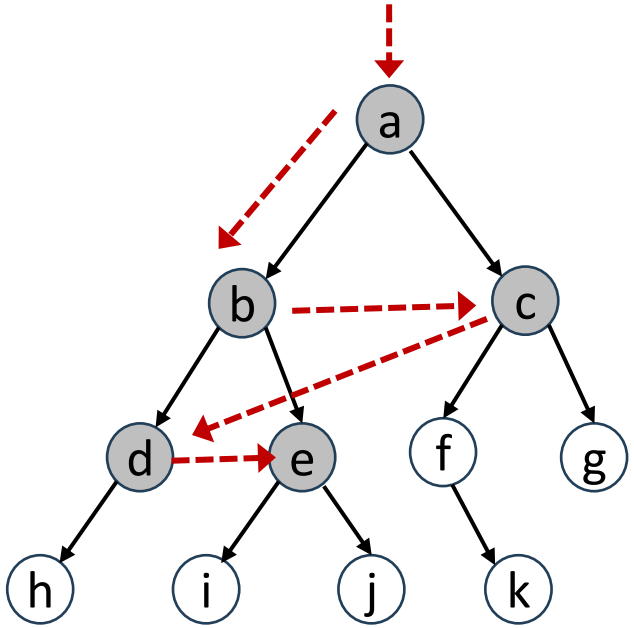

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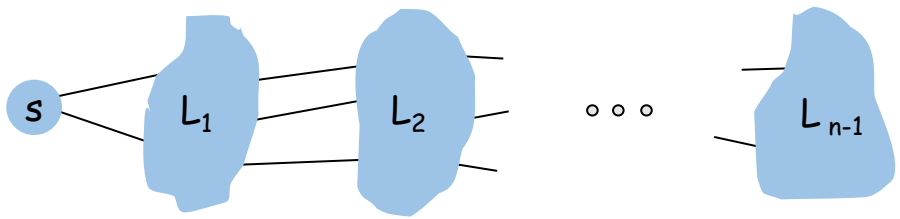

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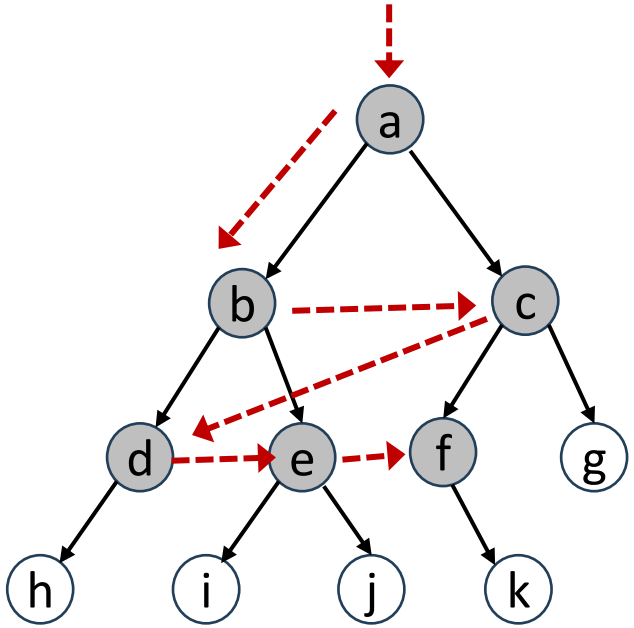

Breath First Search

13

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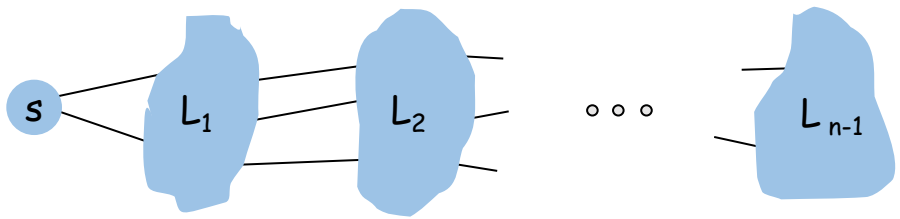

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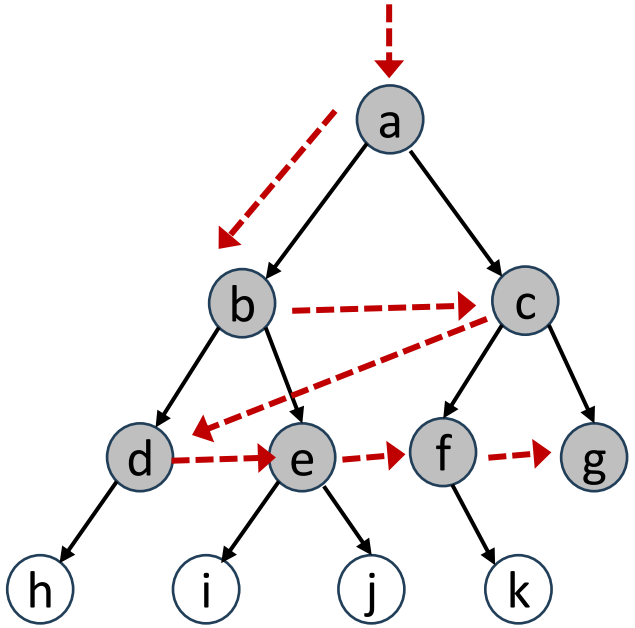

Breath First Search

14

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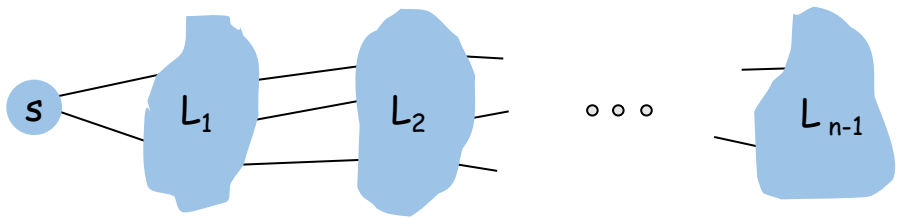

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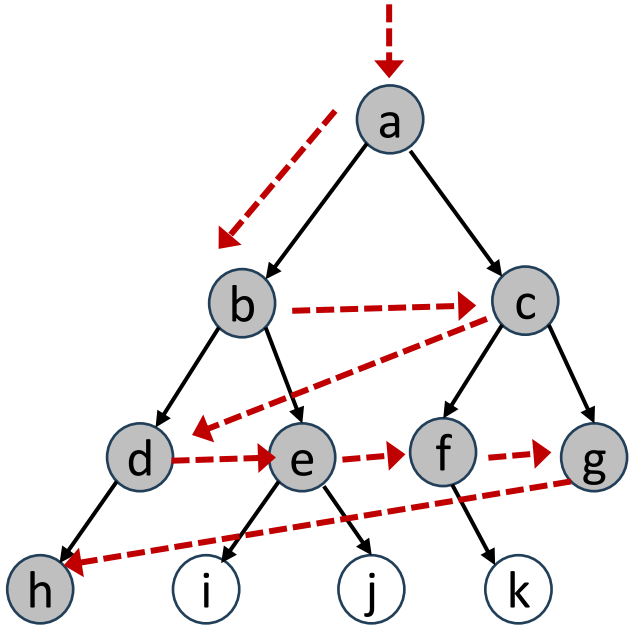

Breath First Search

15

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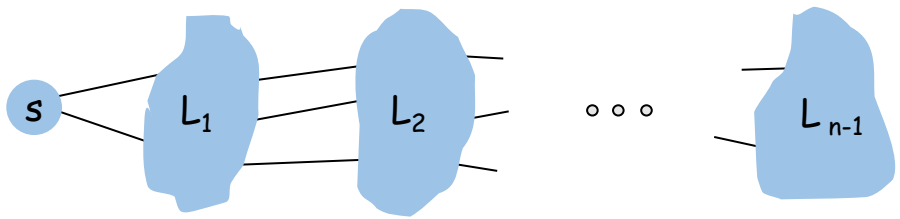

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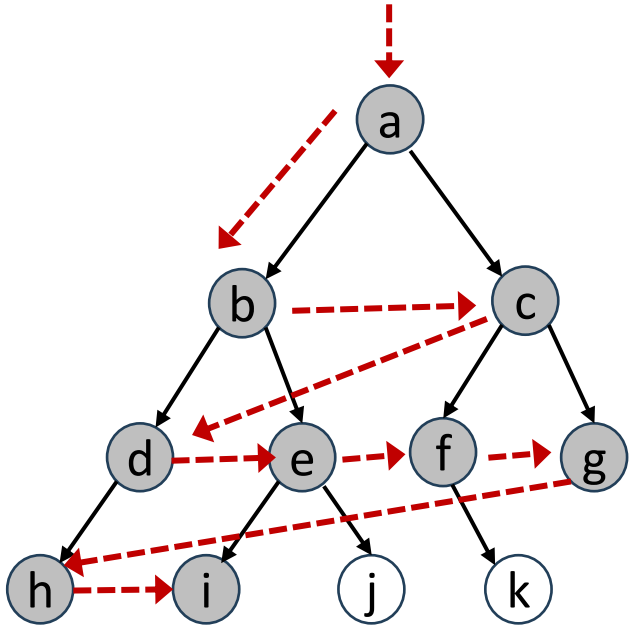

Breath First Search

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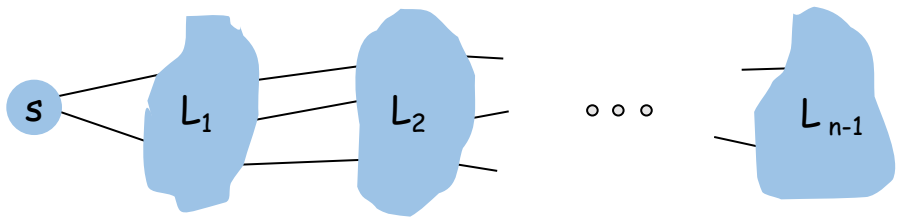

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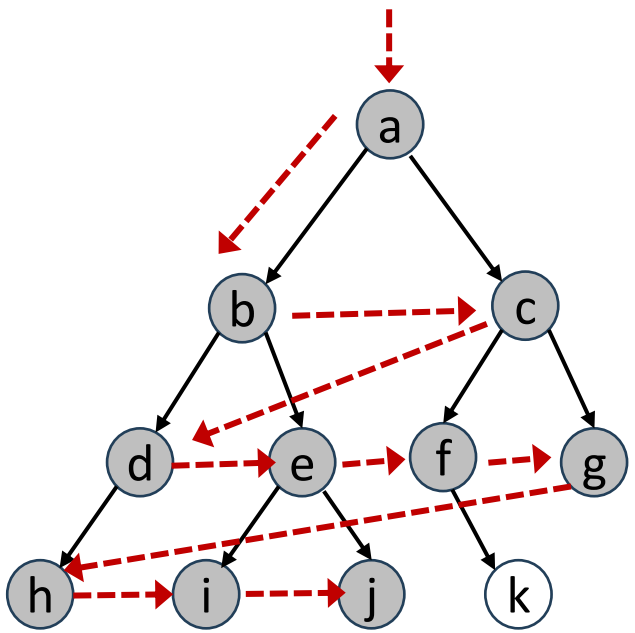

Breath First Search

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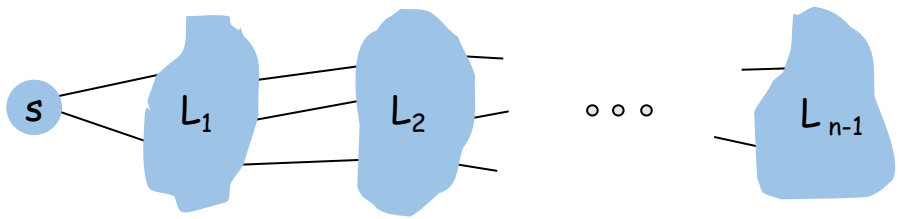

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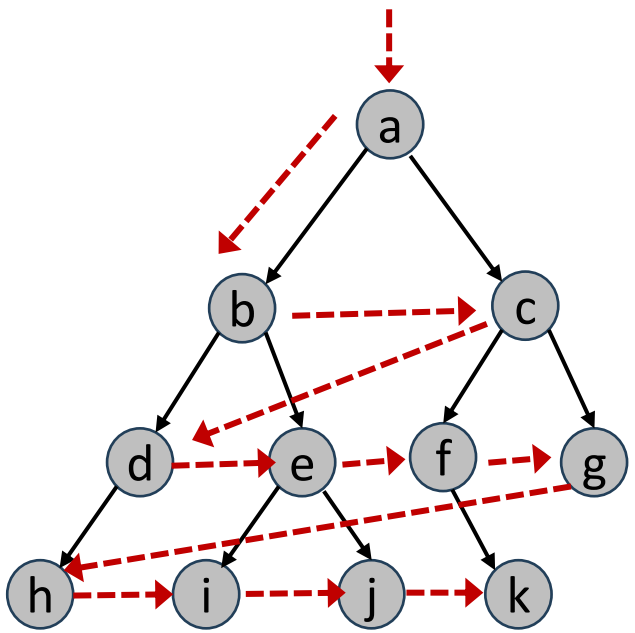

Breath First Search

18

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
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end procedure
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Breadth First Search

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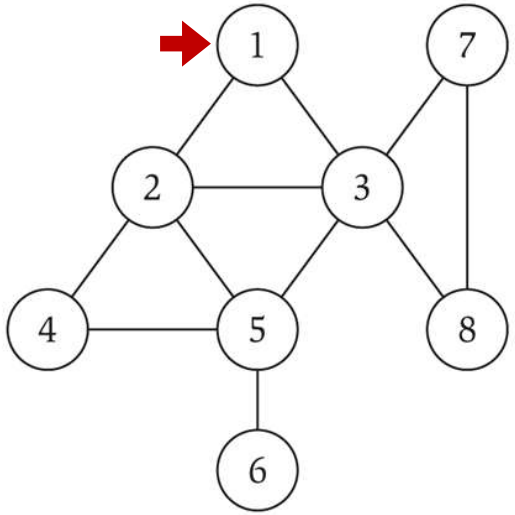

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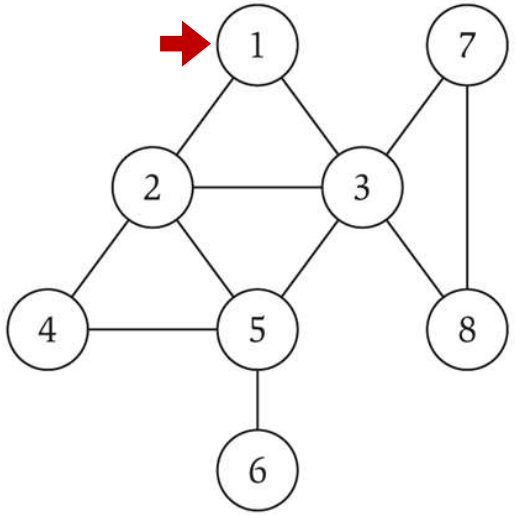


Explored

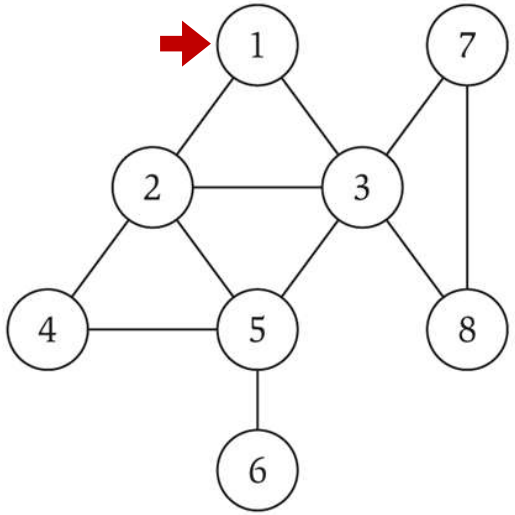
Vertex	Value
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

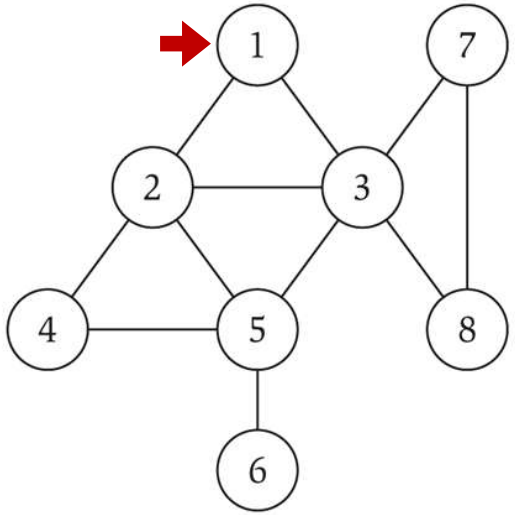
Vertex	Value
1	∞
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

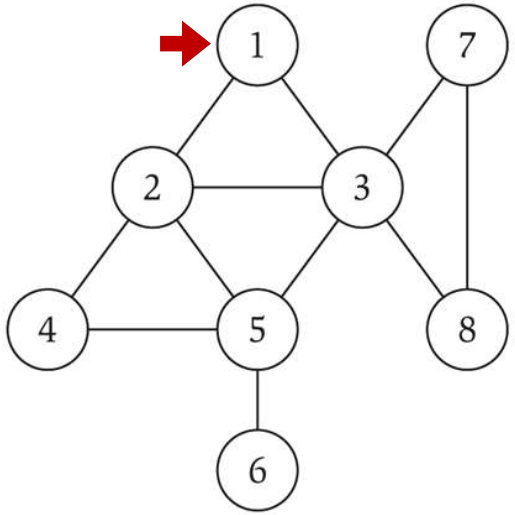
Vertex	Value
1	∞
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

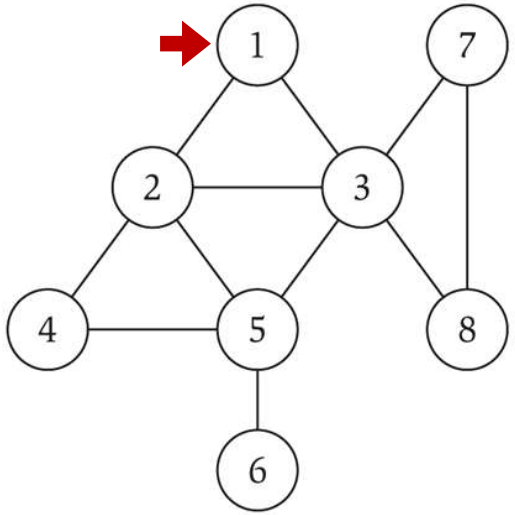
Vertex	Value
1	∞
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

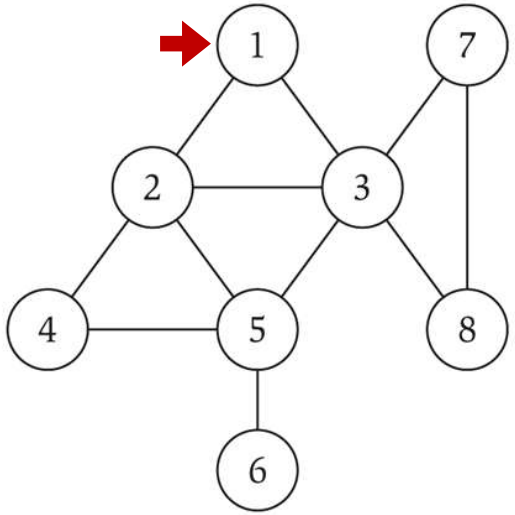
Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

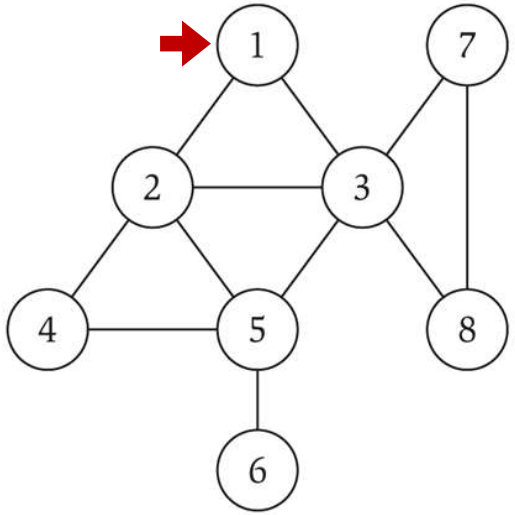
Vertex
1

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
1

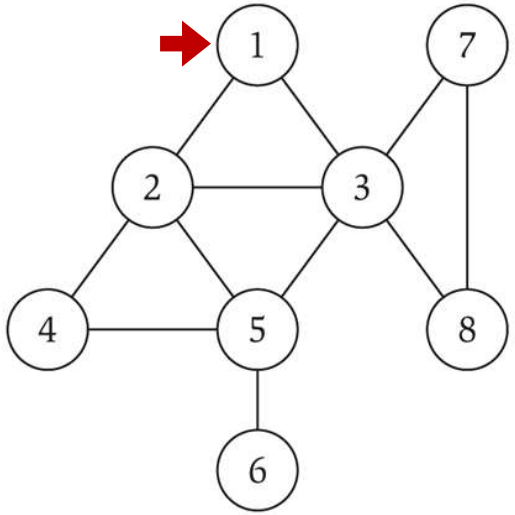
Breadth First Search

U
1

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex

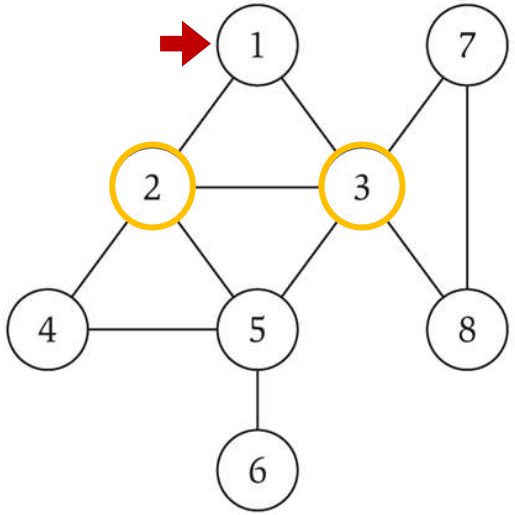
Breadth First Search

U	V
1	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex

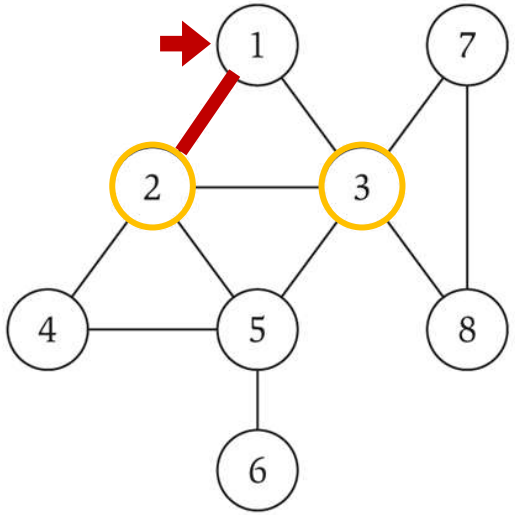
Breadth First Search

U	V
1	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	False
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex

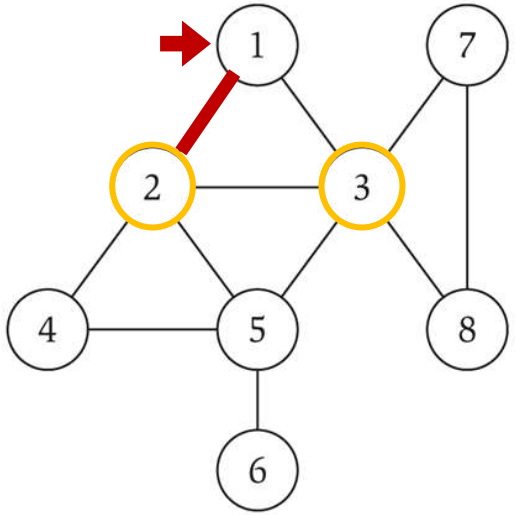
Breadth First Search

U	V
1	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	∞
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex

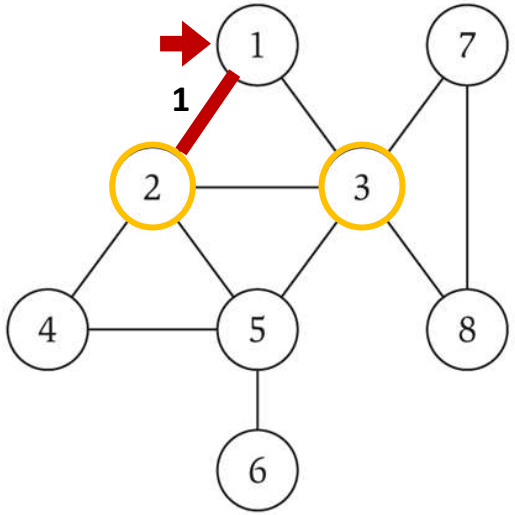
Breadth First Search

U	V
1	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex

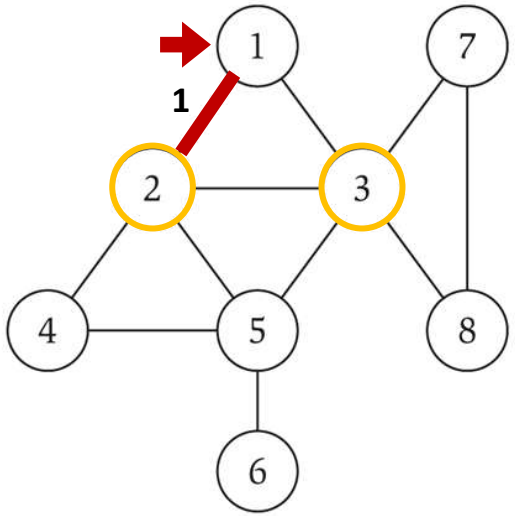
Breadth First Search

U	V
1	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2

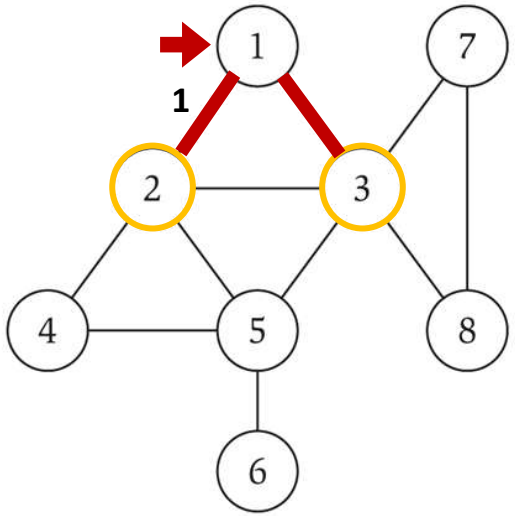
Breadth First Search

U	V
1	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	False
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2

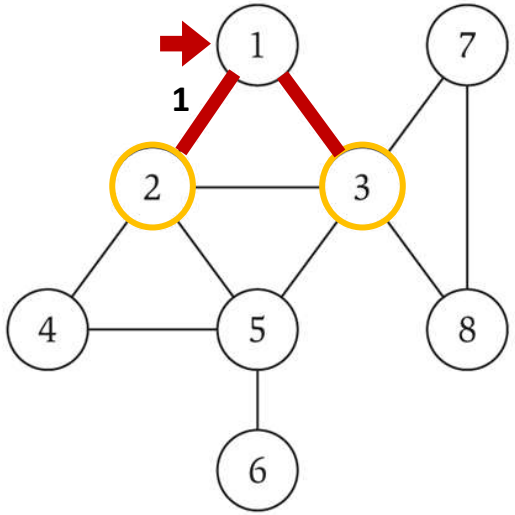
Breadth First Search

U	V
1	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	∞
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2

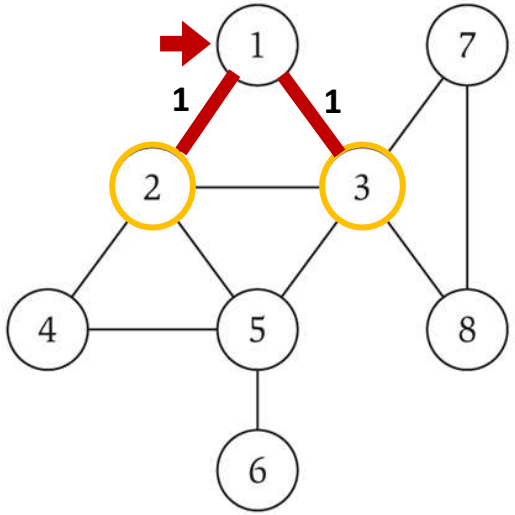
Breadth First Search

U	V
1	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2

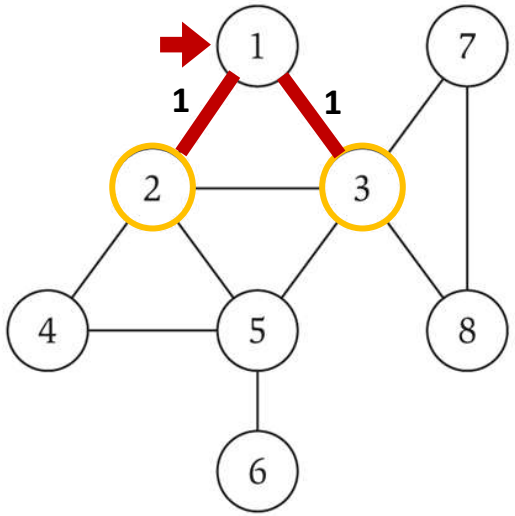
Breadth First Search

U	V
1	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2
3

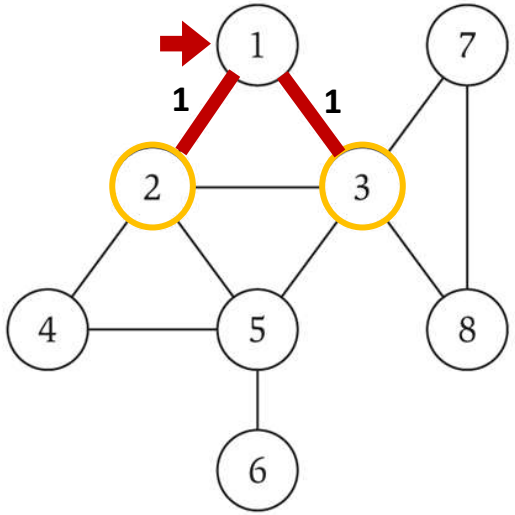
Breadth First Search

U	V
1	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2
3

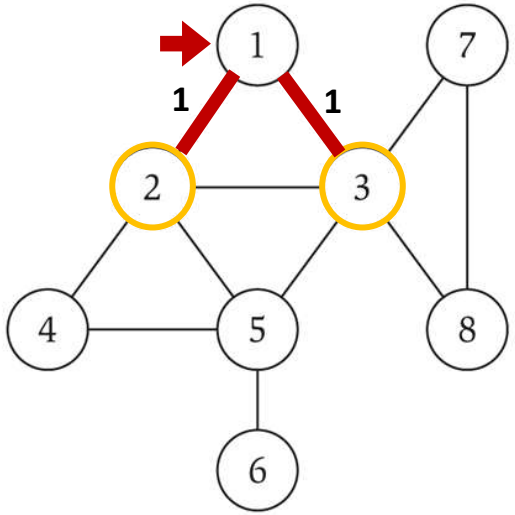
Breadth First Search

U	V
1	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
2
3

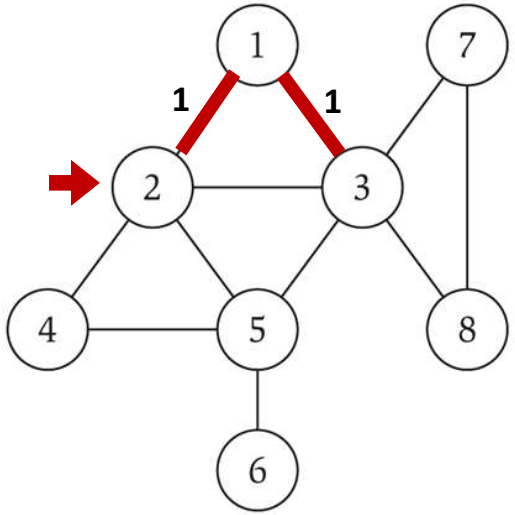
Breadth First Search

U	V
2	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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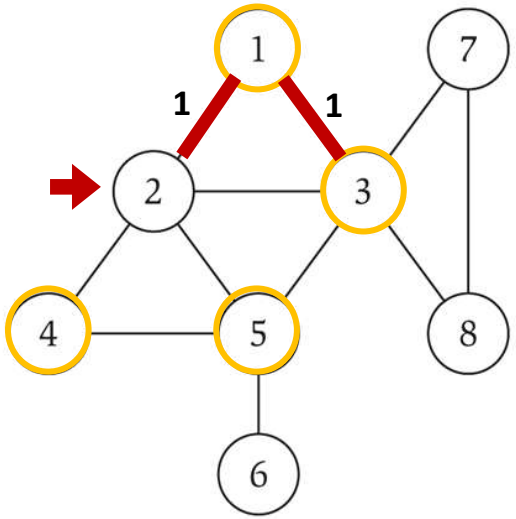
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	1

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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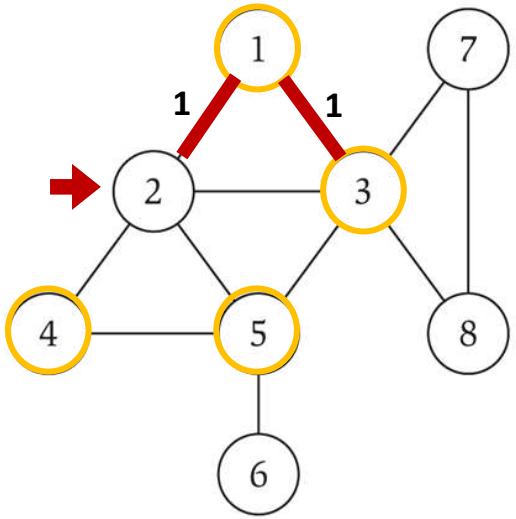
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	1

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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    end for
    explored[s] ← true
    d[s] ← 0
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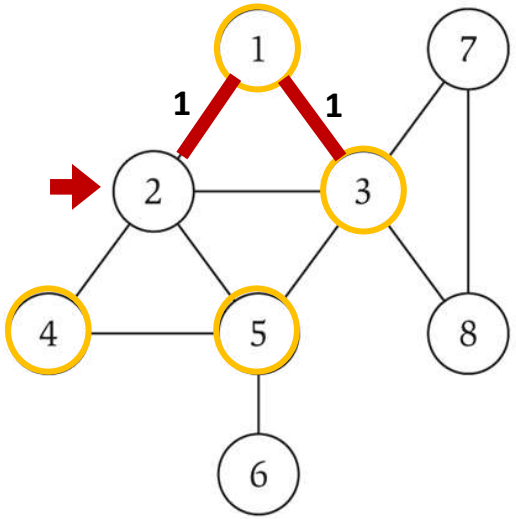
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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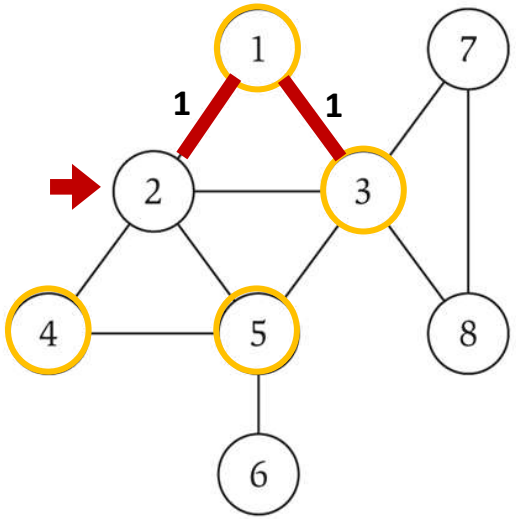
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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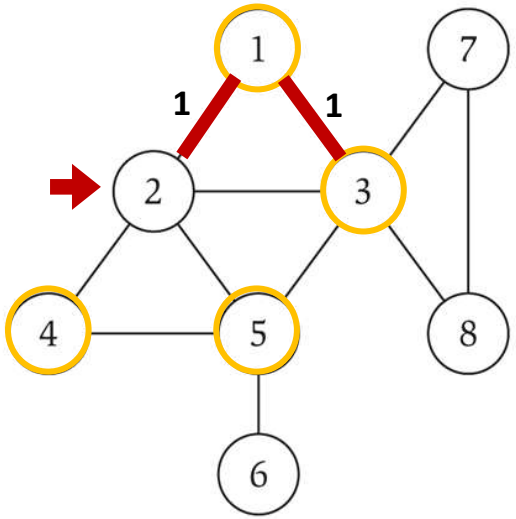
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

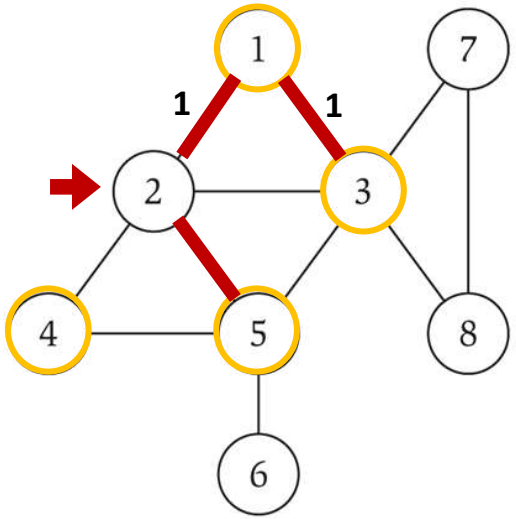
Breadth First Search

U	V
2	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	False
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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        u ← remove vertex from the front of Q
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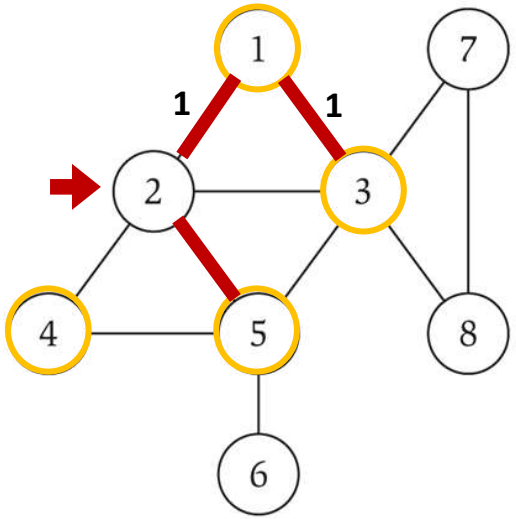
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	∞
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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    explored[s] ← true
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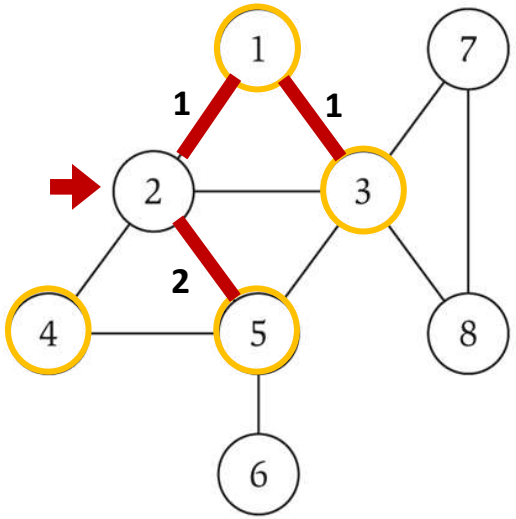
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3

Breadth First Search

U	V
2	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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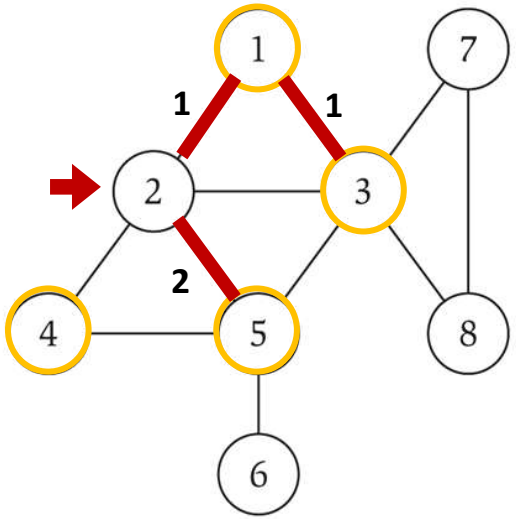
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5

Breadth First Search

U	V
2	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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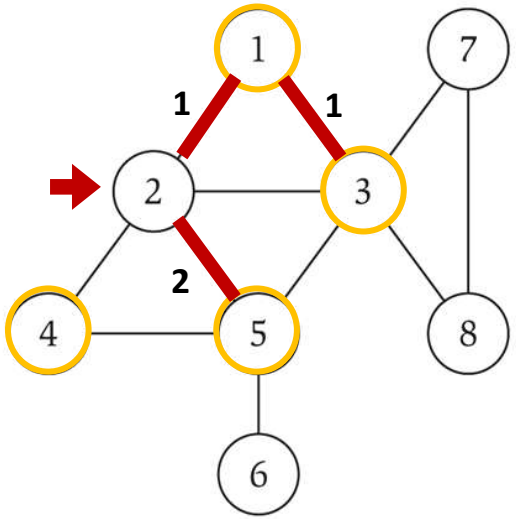
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5

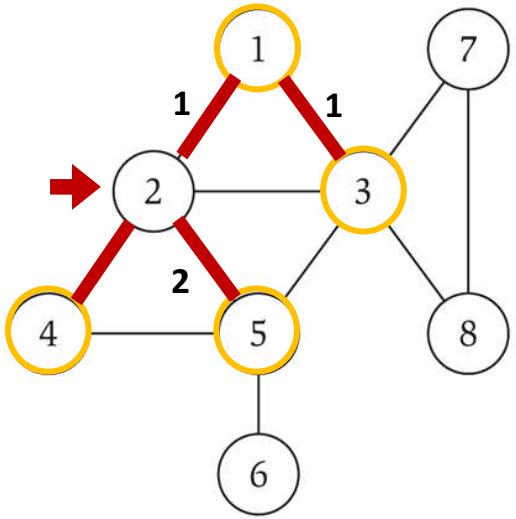
Breadth First Search

U	V
2	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	False
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5

Breadth First Search

U	V
2	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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        u ← remove vertex from the front of Q
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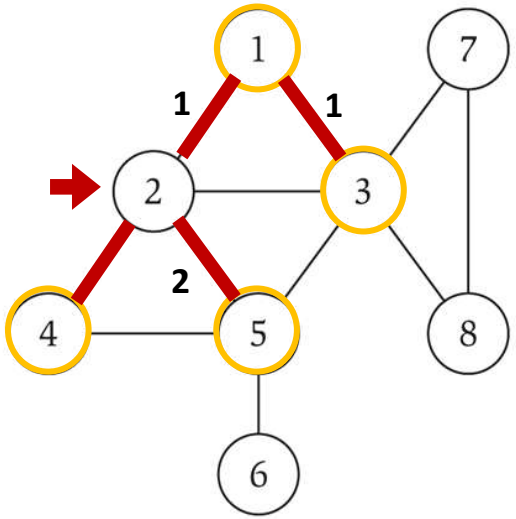
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	∞
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5

Breadth First Search

U	V
2	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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    explored[s] ← true
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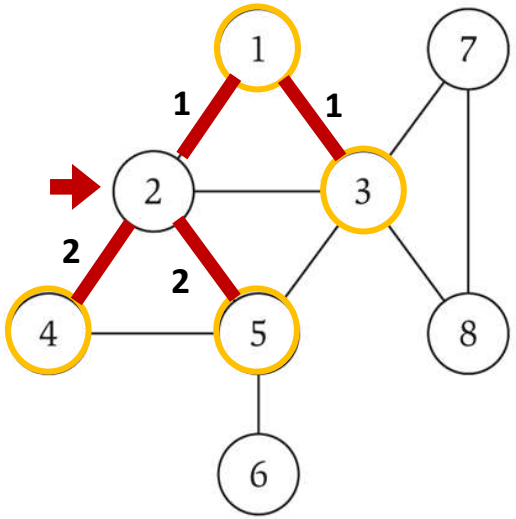
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5

Breadth First Search

U	V
2	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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    explored[s] ← true
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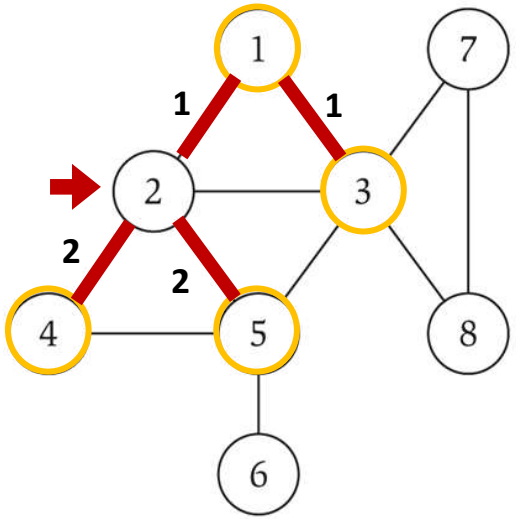
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5
4

Breadth First Search

U	V
2	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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    end for
    explored[s] ← true
    d[s] ← 0
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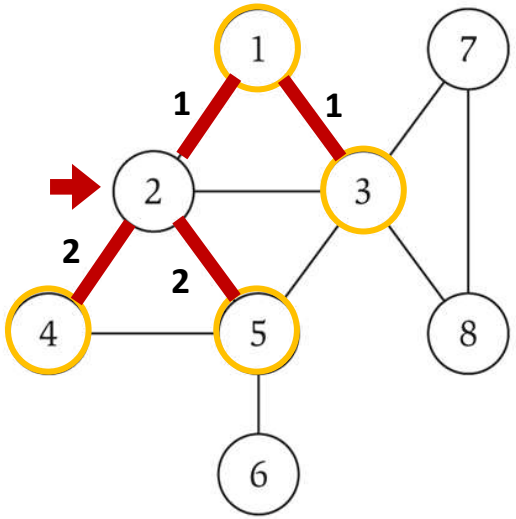
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5
4

Breadth First Search

U	V
2	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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    explored[s] ← true
    d[s] ← 0
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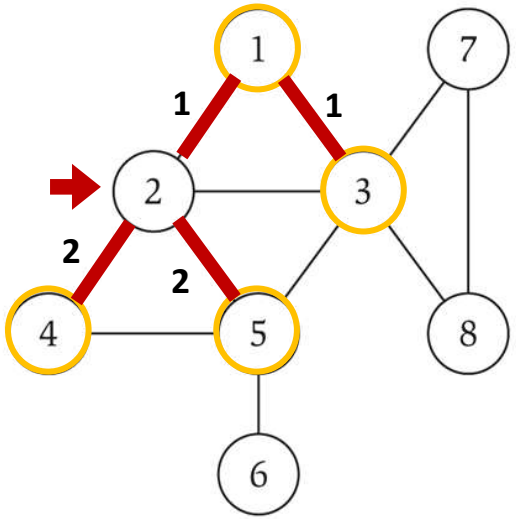
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
3
5
4

Breadth First Search

U	V
3	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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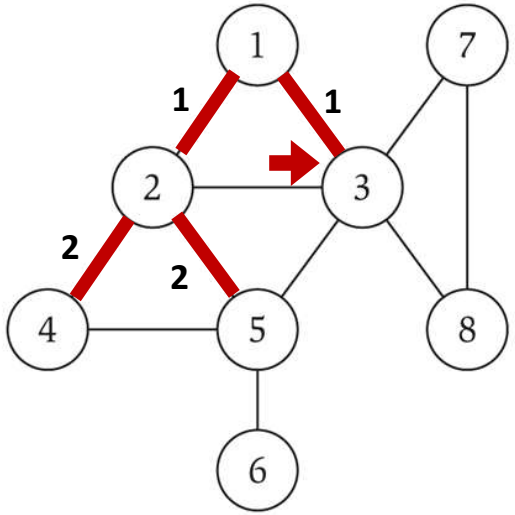
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

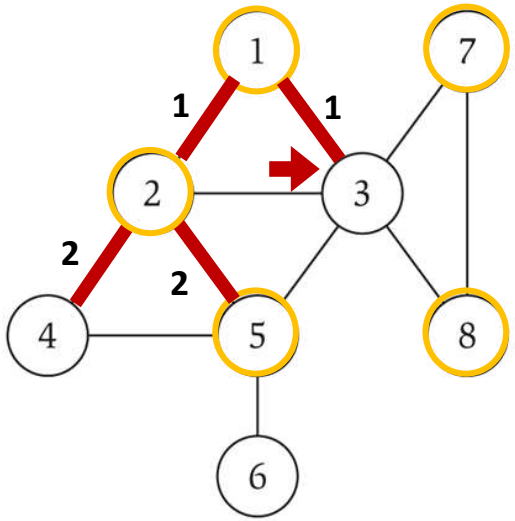
Breadth First Search

U	V
3	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	1

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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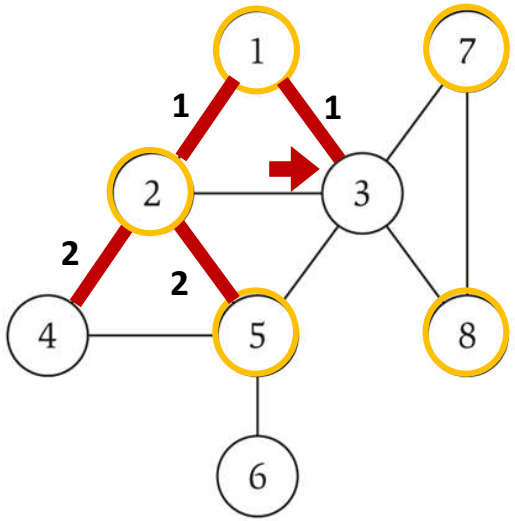
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	1

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
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    explored[s] ← true
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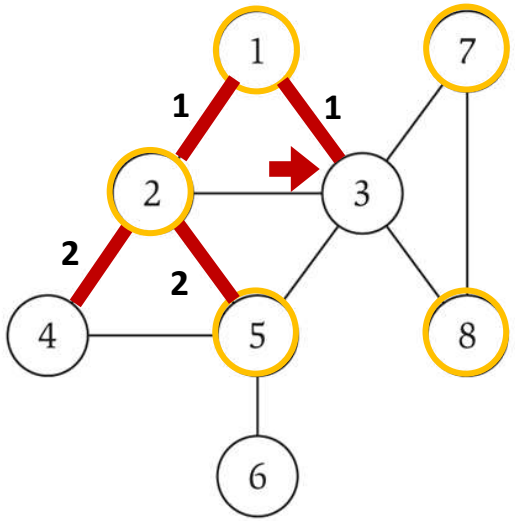
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

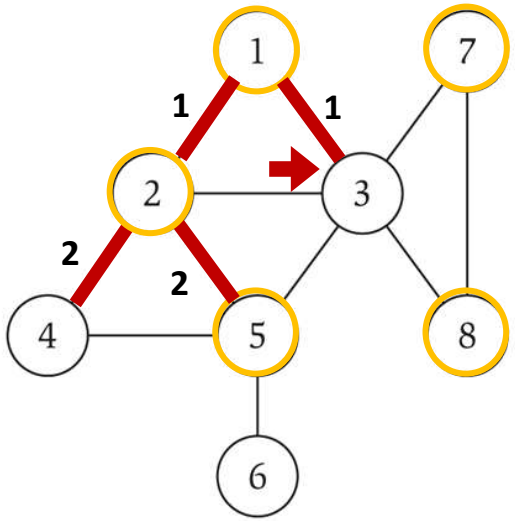
Breadth First Search

U	V
3	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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    end for
    explored[s] ← true
    d[s] ← 0
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                explored[v] ← true
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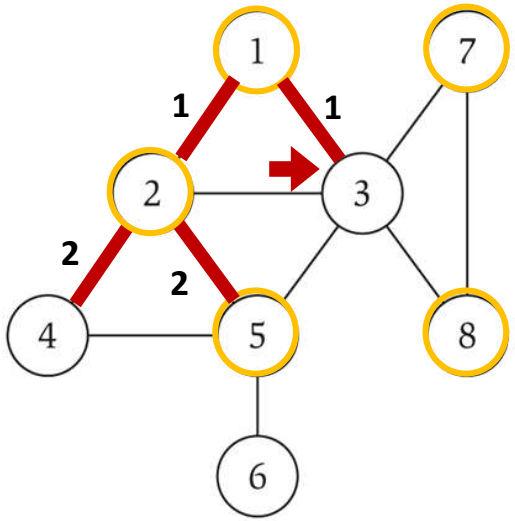
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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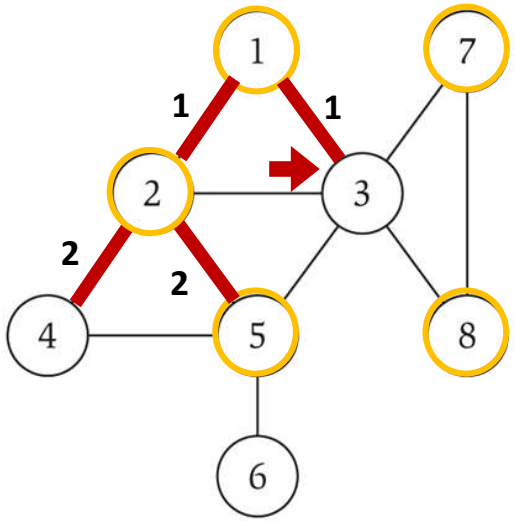
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	5

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
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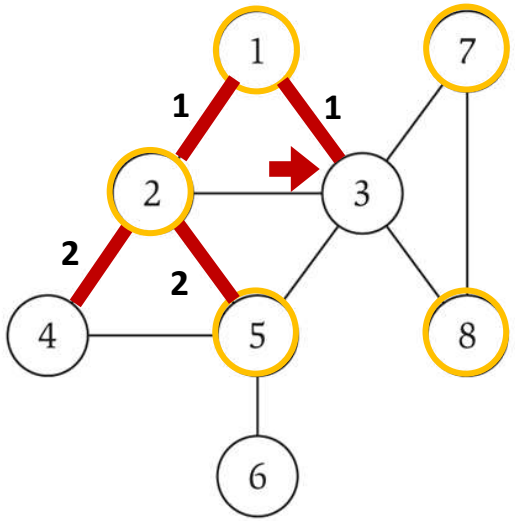
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	8

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
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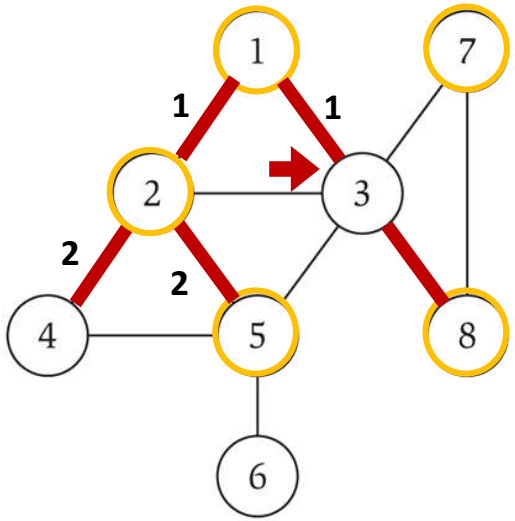
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	False

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	8

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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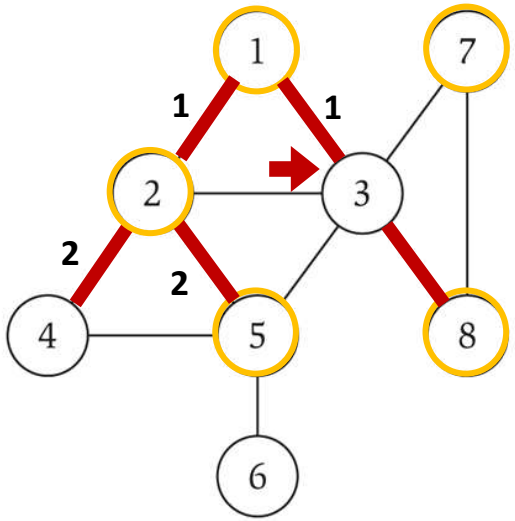
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	∞

Queue

Vertex
5
4

Breadth First Search

U	V
3	8

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procedure BFS(G,s)

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                insert v to the end of Q
            end if
        end for
    end while

end procedure
```

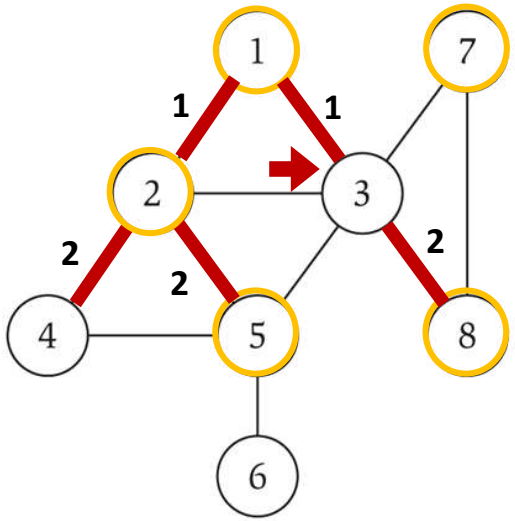

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	2

Queue

Vertex
5
4

Breadth First Search

U	V
3	8

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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                insert v to the end of Q
            end if
        end for
    end while

end procedure
```

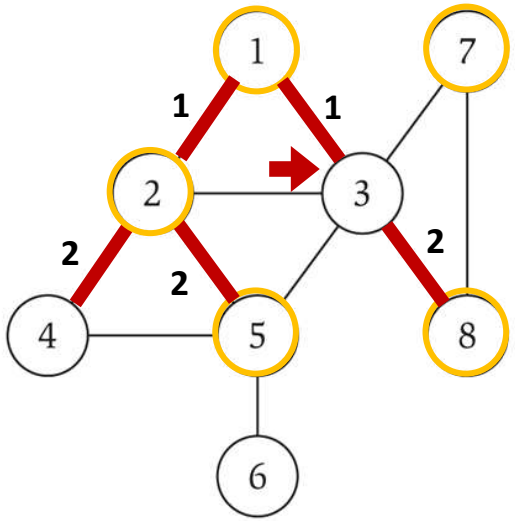

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	2

Queue

Vertex
5
4
8

Breadth First Search

U	V
3	7

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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end procedure
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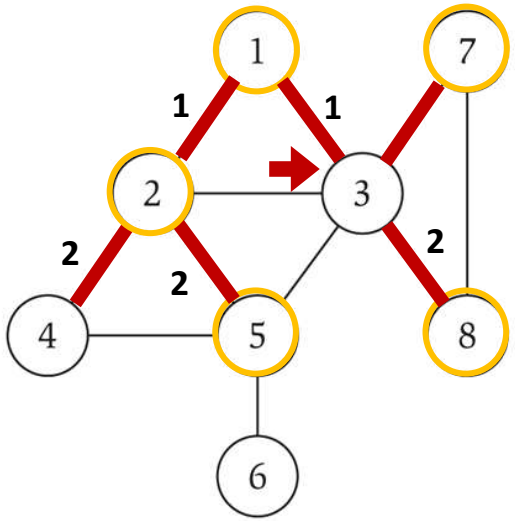

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	False
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	2

Queue

Vertex
5
4
8

Breadth First Search

U	V
3	7

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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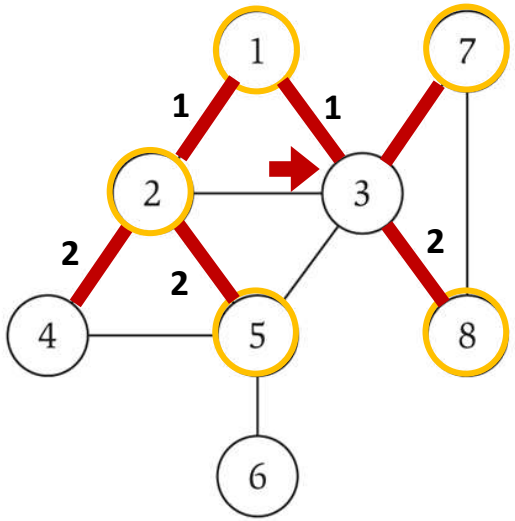
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	∞
8	2

Queue

Vertex
5
4
8

Breadth First Search

U	V
3	7

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end procedure
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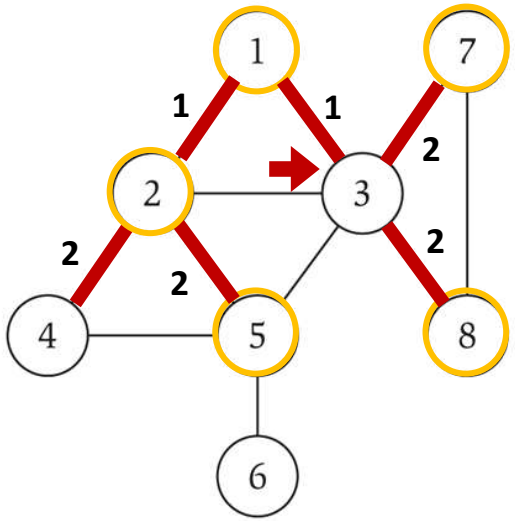

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
5
4
8

Breadth First Search

U	V
3	7

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
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end procedure
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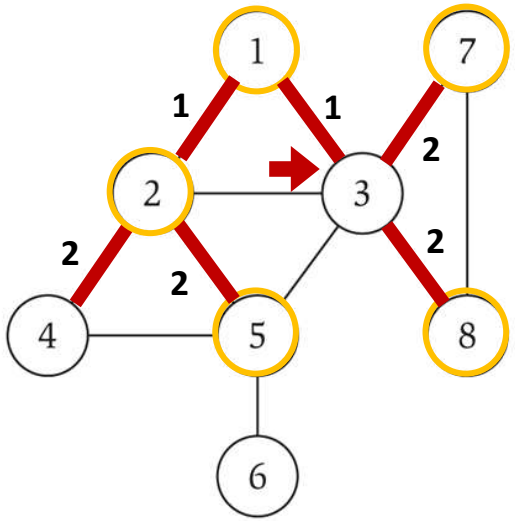

Explored

Vertex	Value
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3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
5
4
8
7

Breadth First Search

U	V
3	7

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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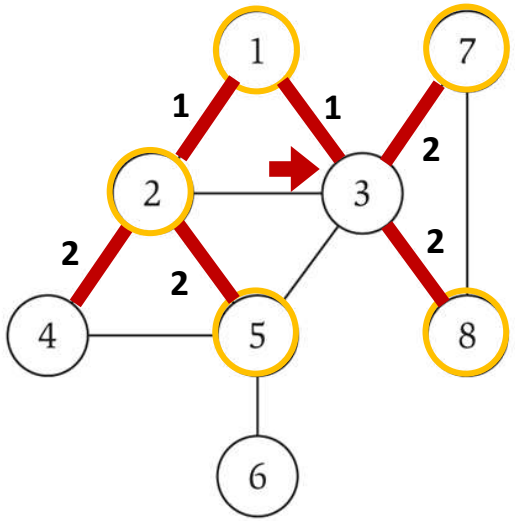

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
5
4
8
7

Breadth First Search

U	V
3	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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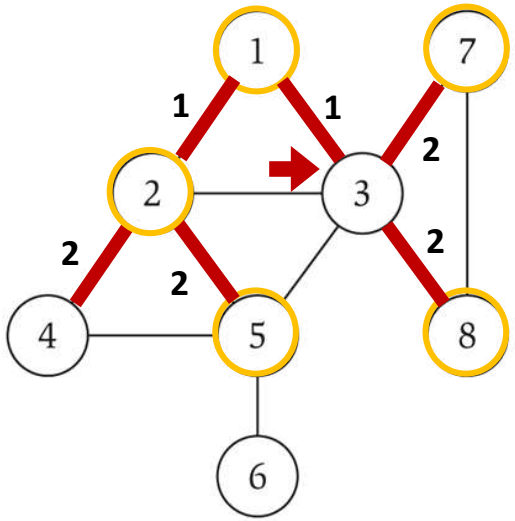
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
5
4
8
7

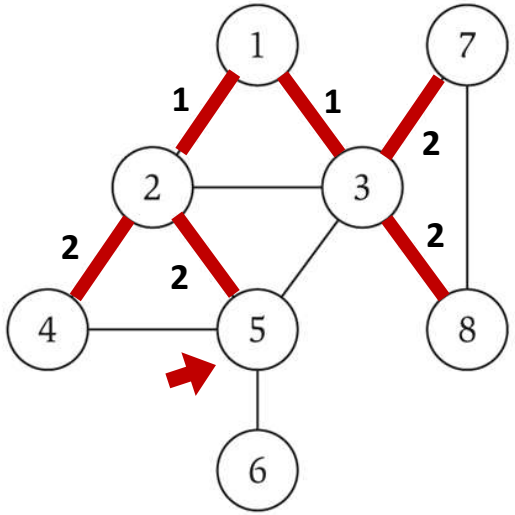
Breadth First Search

U	V
5	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
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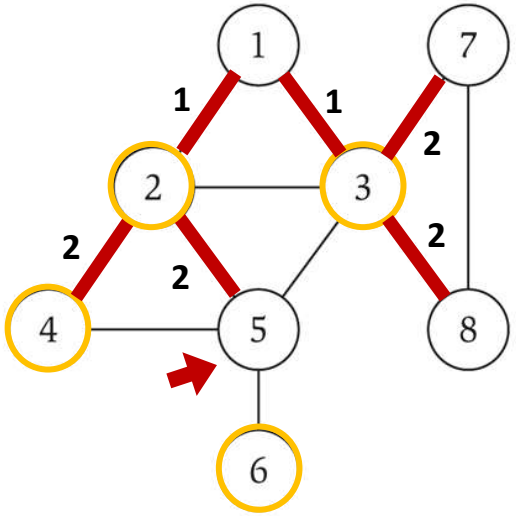
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
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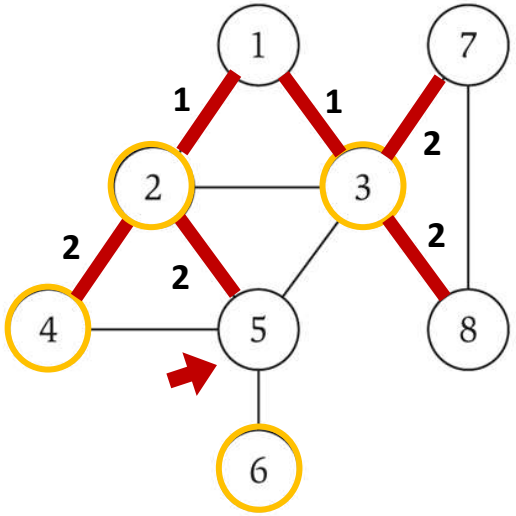
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	4

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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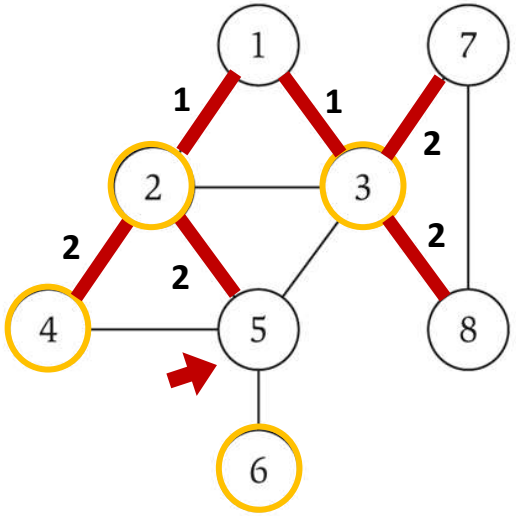
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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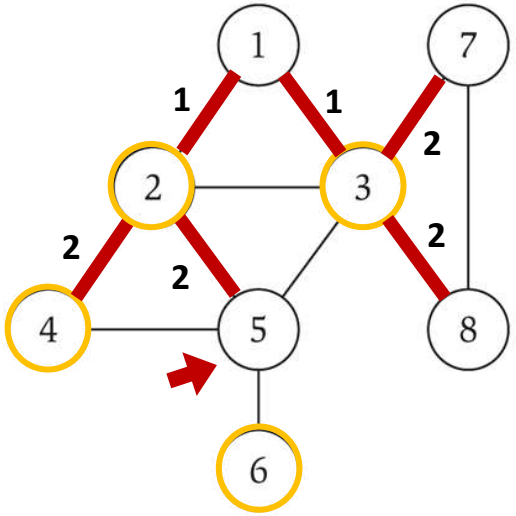
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	2

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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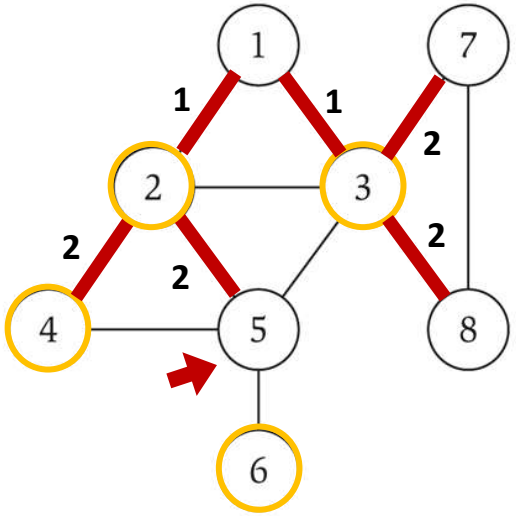
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

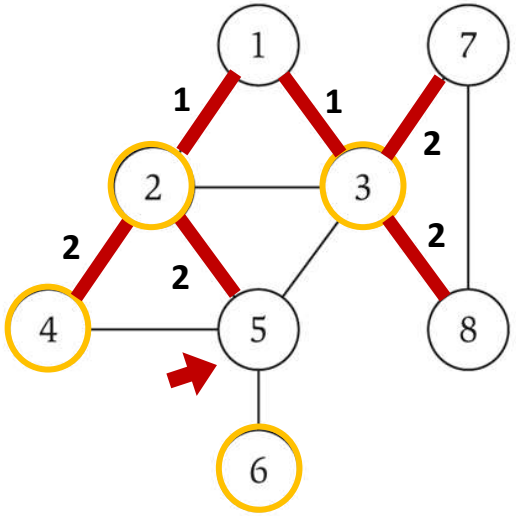
Breadth First Search

U	V
5	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	3

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
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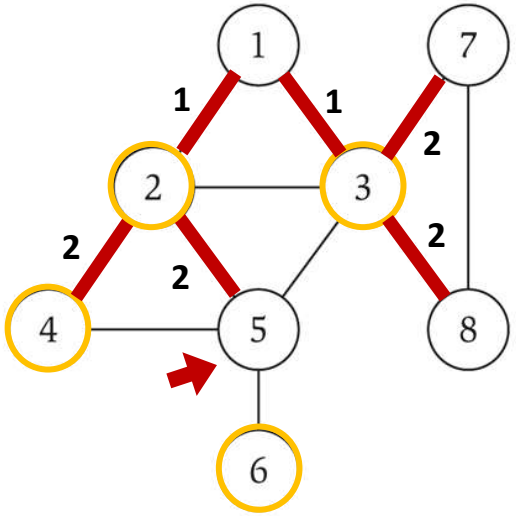
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	8

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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        end for
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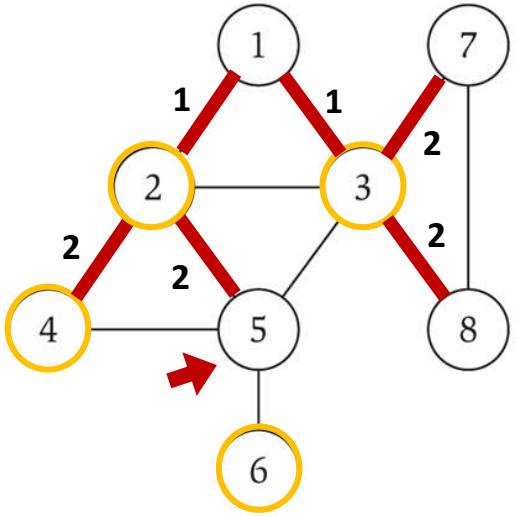
end procedure
```



Explored

Vertex	Value
1	True
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4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	8

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
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    explored[s] ← true
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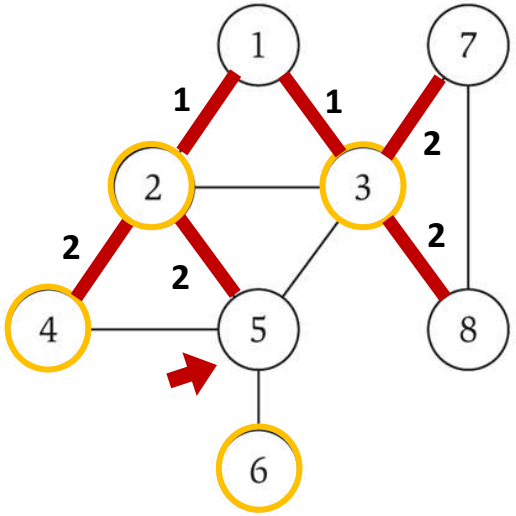
end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	6

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
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    explored[s] ← true
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end procedure
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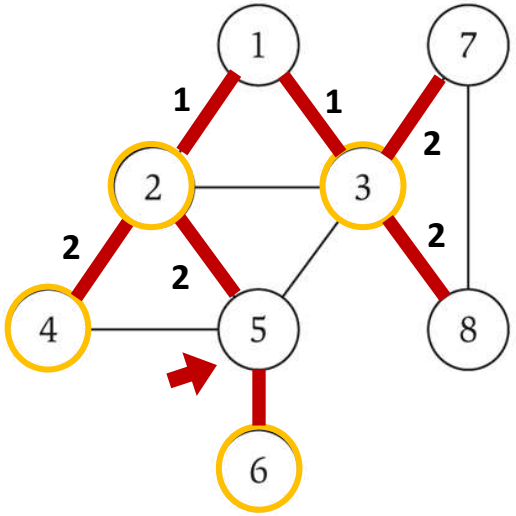

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	False
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	6

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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                insert v to the end of Q
            end if
        end for
    end while

end procedure
```

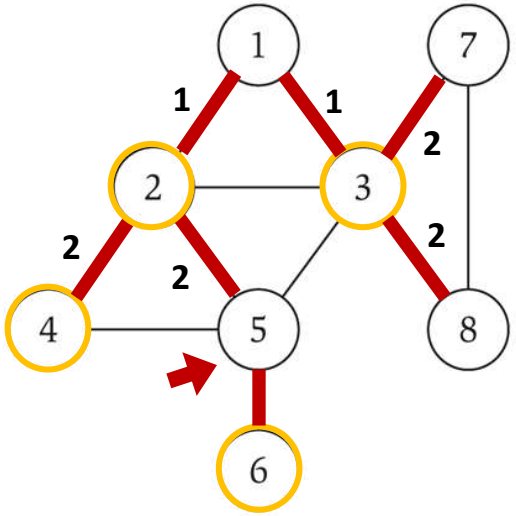

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Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	∞
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	6

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
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            end if
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    end while

end procedure
```

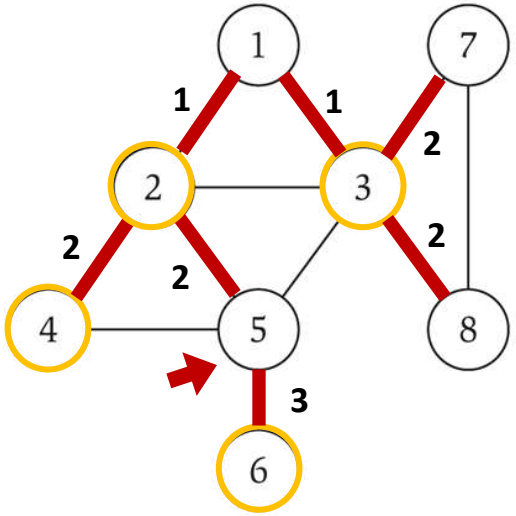

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

U	V
5	6

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
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    explored[s] ← true
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        end for
    end while

end procedure
```

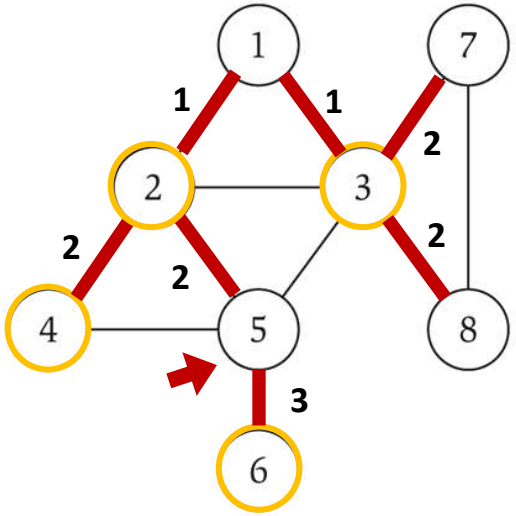

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex
4
8
7
6

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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            end if
        end for
    end while

end procedure
```

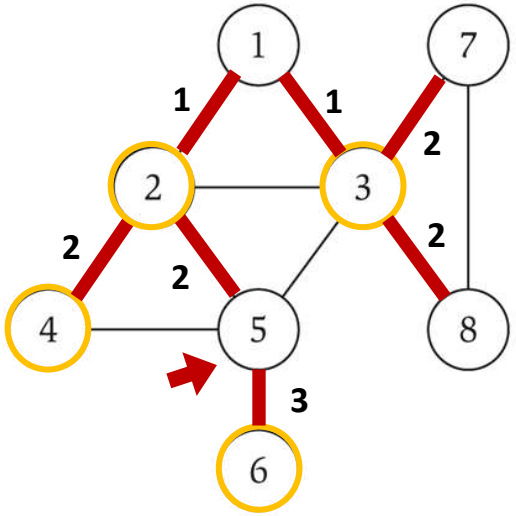

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex
4
8
7

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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    end while

end procedure
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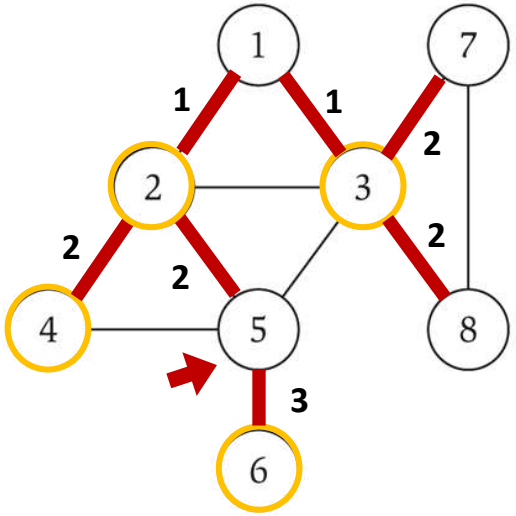

Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex
4
8

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
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                d[v] ← d[u] + 1
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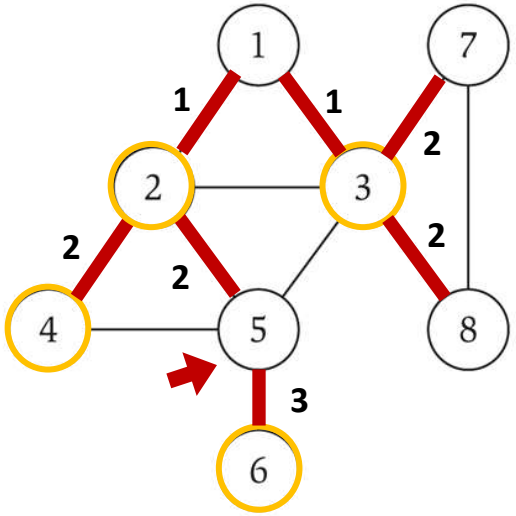
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex
4

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
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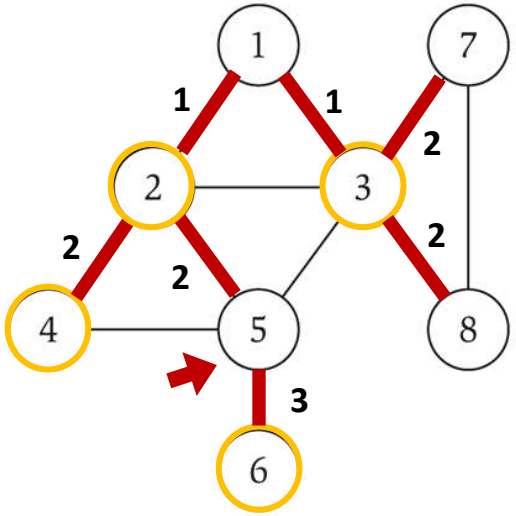


Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

Distance

Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

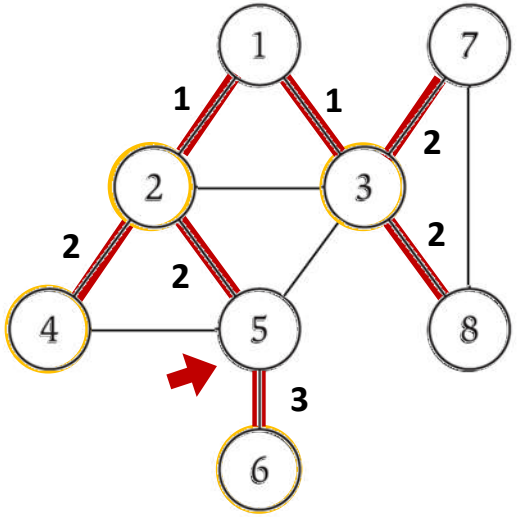
Vertex

Breadth First Search

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```



Explored

Vertex	Value
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True

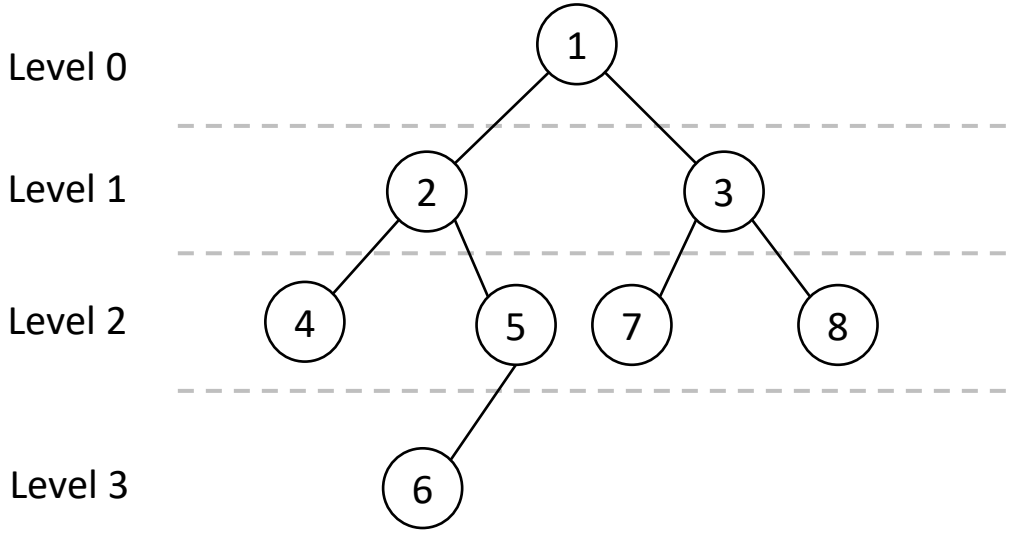
Distance

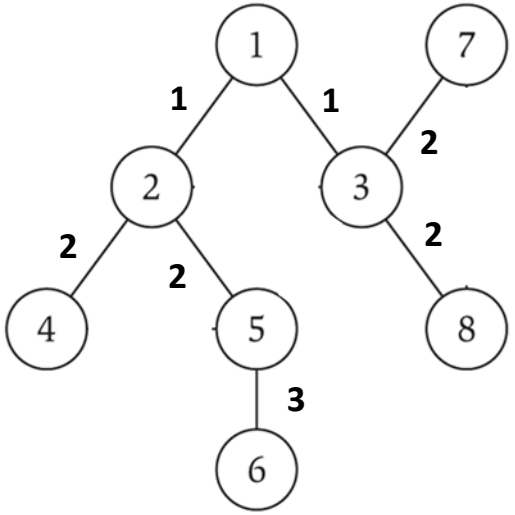
Vertex	Value
1	0
2	1
3	1
4	2
5	2
6	3
7	2
8	2

Queue

Vertex

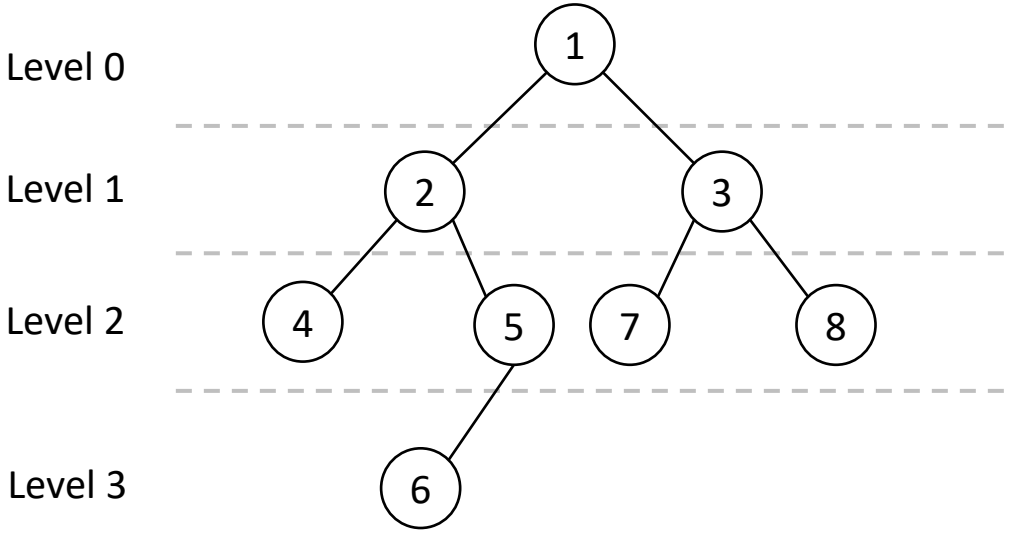
Breadth First Search

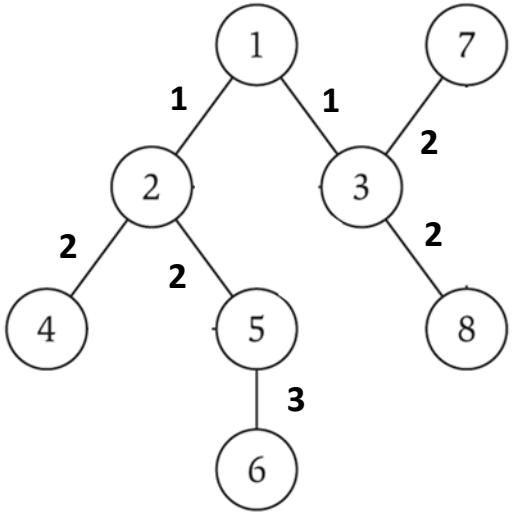




Breadth First Search

Property: Let ***T*** be a BFS tree of **G = (V, E)**, and let **(x, y)** be an edge of **G**. Then the level of ***x*** and ***y*** differ by at most **1**.



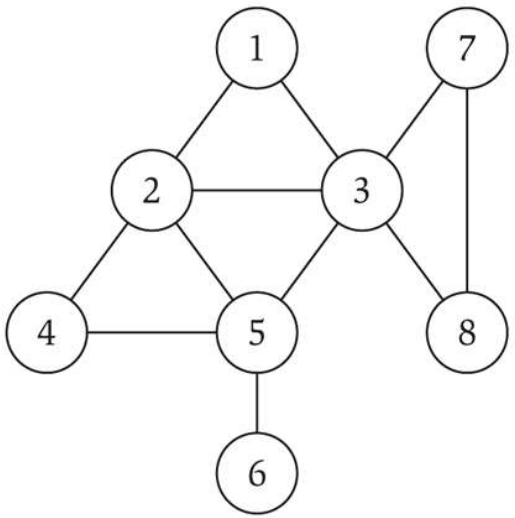


Breadth First Search - Analysis

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false      O(1)
        d[v] ← ∞                 O(1)
    end for                                  O(V)
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for                              O(deg(v))
    end while                                O(V)

end procedure
```

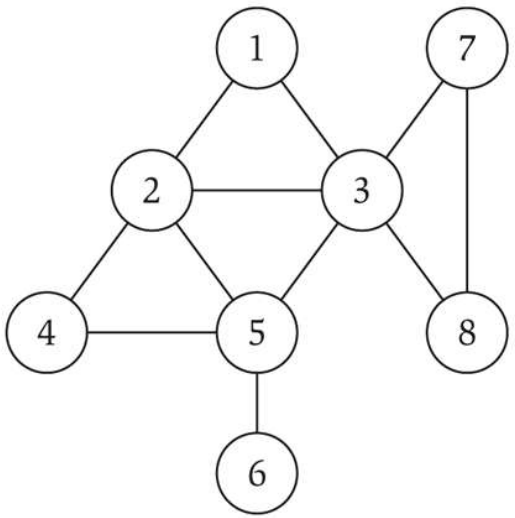


Breadth First Search - Analysis

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false      O(1)
        d[v] ← ∞                 O(1)
    end for                                  O(V)
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while                                O(2E)

end procedure
```



Time Complexity: **O(V) + O(2E)**

O(V + 2E)

O(V + E)

BFS: Live Poll 1

Consider a complete undirected graph where every vertex ***V*** has an edge with every other vertex. You are going to perform a Breadth-First Search (BFS) on this graph.

Which of the following expressions give equivalent time complexity in terms of the Big O notation of the BFS for this graph?

A. V^2

B. 2E

C. V+2E

D. V+E

E. All of Above

F. None of Above

Scan the QR code to vote or go to https://forms.office.com/r/jxW8PBV0Vd

BFS: Live Poll 1

Only people in my organization can respond, Record name

1. Consider a complete undirected graph where every vertex V has an edge with every other...

V^2	18%
2E	4%
V+2E	16%
V+E	16%
All of Above	45%

51 responses

1/1

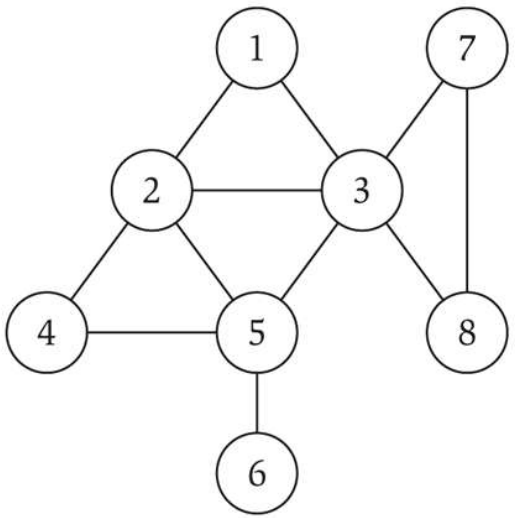
Scan the QR code to vote or go to https://forms.office.com/r/jxW8PBV0Vd

Breadth First Search - Analysis

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false      O(1)
        d[v] ← ∞                 O(1)
    end for                                  O(V)
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for                              (V-1) = O(V)
    end while                                O(V)

end procedure
```



Time Complexity: $O(V) + O(V^2)$

$O(V^2)$

BFS: Live Poll 1

Consider a complete undirected graph where every vertex ***V*** has an edge with every other vertex. You are going to perform a Breadth-First Search (BFS) on this graph.

Which of the following expressions give equivalent time complexity in terms of the Big O notation of the BFS for this graph?

A. $V^2 = \mathbf{O(V^2)}$ ✓

B. 2E = $V(V-1) = V^2 - V = \mathbf{O(V^2)}$ ✓ 😈

C. V+2E = $V + V(V-1) = V^2 = \mathbf{O(V^2)}$ ✓ 😈

D. V+E = $V + (V(V-1))/2 = V + (V^2 - V)/2 = \mathbf{O(V^2)}$ ✓ 😈

E. All of Above

F. None of Above

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over